Frozen VLA at Test Time



- Fixed weights (0.4-3.6B params)
- No task-specific fine-tuning
- Variable action quality

Adaptive Conformal Refinement

VLA → K=2 Sample → Adaptive Gate → CRM → a^*

615K params | 1.8 MB | ~170ms latency

Adapts intervention to VLA uncertainty

Selective Improvement

Spatial: ▲+2.1pp	Goal: ▲+1.3pp
Object: −+0.1pp	Long: −-0.3pp

"Improves where uncertain, preserves where confident"



High uncertainty -> CRM active

Gating correctly modulates intervention intensity



Low uncertainty -> CRM bypassed

Model Size

615K

parameters (1.8 MB)

Latency

~170ms

all K samples batched

Best Improvement

+2.1pp

Spatial (7.8% trigger)

Cross-Model

2 VLAs

SmolVLA + pi0.5 validated